# Bess Hagan

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## Summary

Computer Science graduate (4.0 GPA) with hands-on experience building a SQL-to-Python data pipeline and transforming multi-table relational data into structured, analysis-ready formats. Skilled in cleaning, validating, and analyzing complex datasets to ensure accuracy and reliability. Coauthored a peer-reviewed publication presented at the 21st AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE-25).

## Education

**B.A. Computer Science | Minor in Data Analytics** *Dec 2025*

Southwestern University, Georgetown, TX

- GPA: 4.0, Summa Cum Laude
- Chapter President, Upsilon Pi Epsilon International Honorary Computer Science Society (2024–2025)
- **Selected Coursework:** Database Management, Data Analytics, Statistics, Machine Learning

## Experience

**E-Commerce Sales Analyst** *Feb 2026 – Mar 2026*

Independent Project

- Analyzed ~14,000 e-commerce transactions to identify revenue drivers and purchasing behavior patterns, addressing five defined business questions.
- Conducted full analysis in Excel (Pivot Tables, charts) then reproduced in Python with Pandas and Matplotlib to validate results and assess workflow trade-offs.
- Identified key revenue drivers through analysis and developed clear, actionable insights to support business decision making, recommending retention-focused growth strategies over short-term tactical optimizations.
- Built structured reports and visualizations to communicate trends and insights to non-technical audiences.

**Procedural Content Generation Researcher** *May 2025 – Jul 2025*

Southwestern University (SURF 2025), Georgetown, TX

- Coauthored a peer-reviewed paper presented at the 21st AAAI Conference on AI and Interactive Digital Entertainment (AIIDE-25).
- Trained conditional and unconditional diffusion models with UNet architecture in PyTorch for image-based procedural game level generation; implemented cross-entropy loss and early stopping within an existing codebase.
- Developed Python scripts to evaluate and visualize A* solvability metrics across models, and created batch scripts for data preparation, training, and results reporting.

**Data Researcher** *May 2024 – Aug 2024*

University of Illinois Urbana-Champaign (INVITE Institute), Champaign, IL

- Analyzed structured behavioral log data using SQL and Python across ~100 users (62 accounts) in a Minecraft-based learning environment to identify markers of student persistence and struggle.
- Built a SQL-to-Python data pipeline to extract, clean, and transform multi-table relational event log data into structured, analysis-ready binary datasets.
- Defined and implemented 10+ behavioral event types to transform raw log data (e.g., puzzle completion state, run and reset event counts, Levenshtein edit distance between solution attempts) into structured, analysis-ready binary datasets.
- Performed data validation and consistency checks across datasets to ensure accuracy and reliability for downstream analysis and reported cross-system discrepancies to developers.

## Skills

**Data Analysis & Visualization:** Python (Pandas, NumPy, Matplotlib), SQL, R, Excel (Pivot Tables, Charts), Microsoft Power BI, Exploratory Data Analysis (EDA), Data Cleaning & Transformation, Data Modeling

**Databases & Tools:** PostgreSQL, pgAdmin, Data Pipelines, Jupyter Notebook, Git/GitHub, VS Code, Anaconda

**Machine Learning:** scikit-learn, PyTorch, Linear & Logistic Regression, Decision Trees, KNN, K-Means